

# The Root Cause, Verified: Smoothness and the Regime Dial

The root-cause analysis of the active-set bridge - "the wrong order of smoothness" - evaluated claim by claim against the executed M1/M3/M4a/M4b record, corrected where it misdescribes the measured objects, and extended by a new measurement, M4c, which gives its "separate regime" a mechanism and a law: one curvature measure, multiple resolutions, a measurable crossover.

## Computational Verification Unit

Source: the root-cause analysis under evaluation · verdicts grounded in the v2 theorem set (S/G/N1/N/D), the committed M4a/M4b artifacts, and 858 new lexicographic solves at the M4b codim-2 vertex · frozen v21 untouched

The diagnosis is correct and theorem-backed; three misdescriptions are corrected (no  $O(\epsilon)$  holonomy exists - the scalings form a trichotomy; the slope-1.00 statistic is the 9-of-76 interacting stratum; the slope-1 law belongs to the dynamic layer). Theorem R closes the gap: the smoothed map's curvature is the same measure, its epsilon-squared law measures at slope 1.9991-1.9995, and the crossover scales linearly, epsilon-star approximately 3 sigma. The unification is a resolution statement, not a limit statement.